HHS3190M
HUMAN PHYSIOLOGY AND FUNCTIONAL ANATOMY FOR REHABILITATION SERVICES
HHS3190MJ
復康服務之人體生理學及功能性解剖學

Physiology L2: Human Cells, Basic Tissues, Organs

生理學 L2：人體細胞、基礎組織與器官

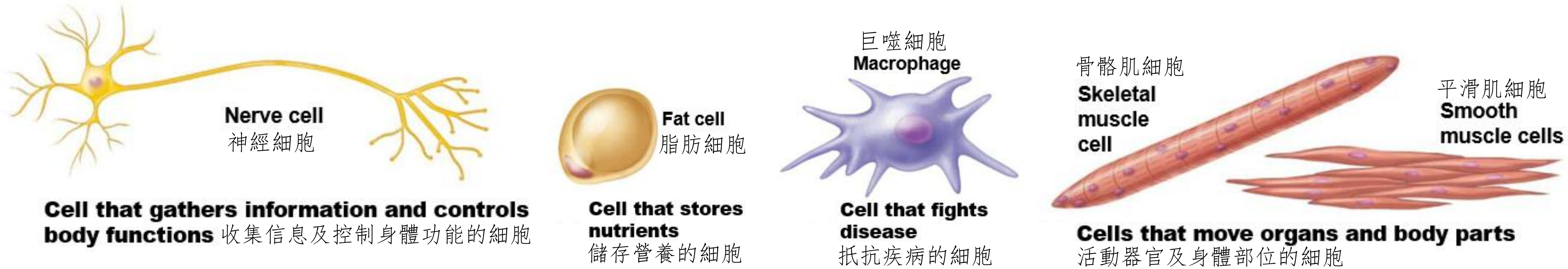
(Sem 1 AY26-27 26-27學年第1學期)

Outline大綱

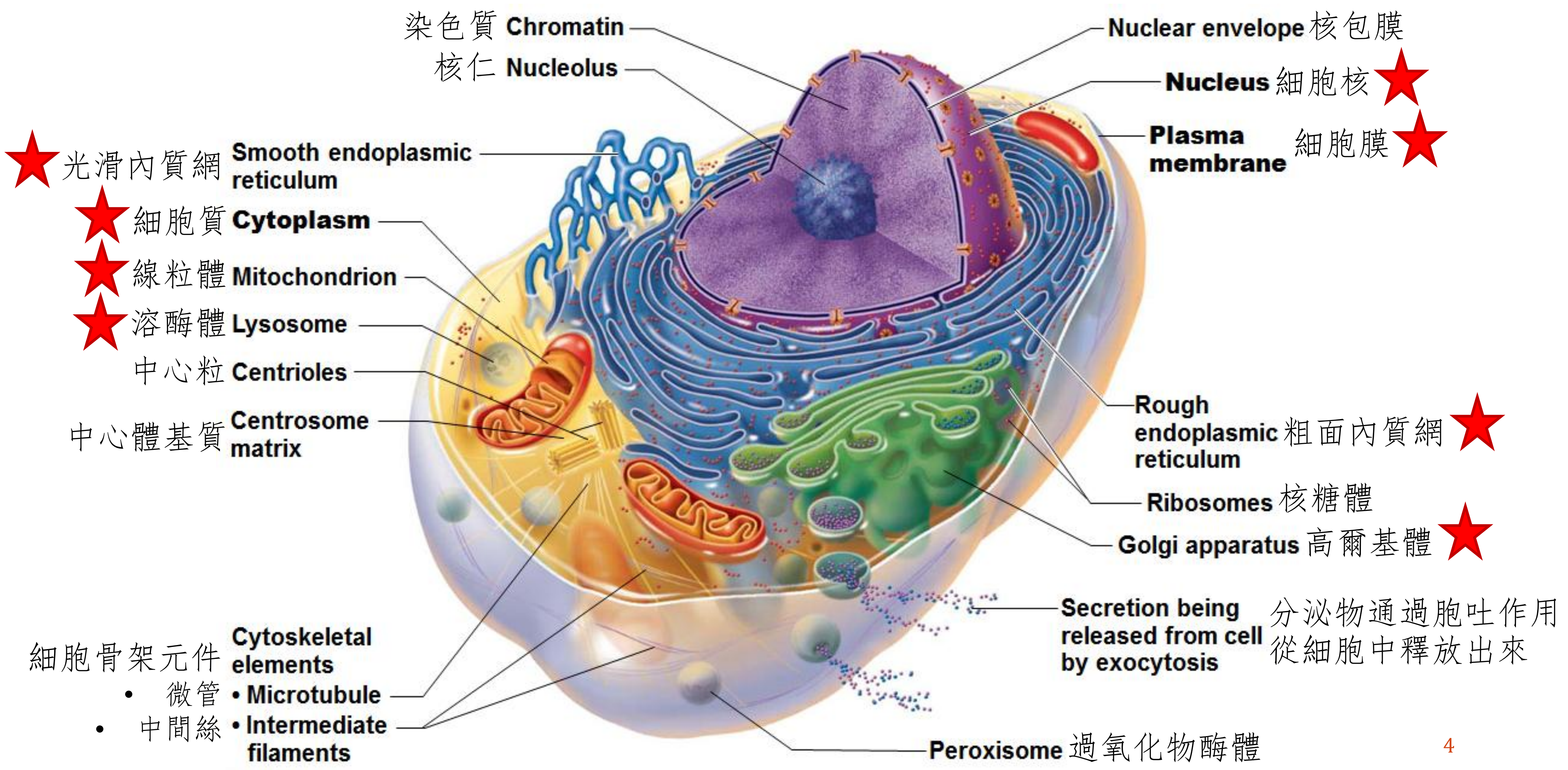
- Cell Organelles in a Typical Human Cell
  典型人類細胞中的細胞器
- Primary Tissues in Human Body and their Significance
  人體基礎組織及其重要性

Typical Human Cell典型人類細胞

- Cell theory 細胞理論
- A **cell** is the **structural and functional unit** of life. All living organisms are composed of cells **細胞**是生命的**結構和功能單位**，所有生物都由細胞組成。
- Human is **multicellular organism** 人類是**多細胞生物**。
  - Human body is composed of many cells 人體由許多細胞組成
  - Over 200 different types of human cells 超過 200 種不同類型的人體細胞
  - Types differ in size, shape, and subcellular components; these differences lead to differences in functions
    不同類型在大小、形狀和細胞構成部份方面有所不同;這些差異導致了功能上的差異



Cell Organelles in a Typical Human Cell 典型人類細胞中的細胞器



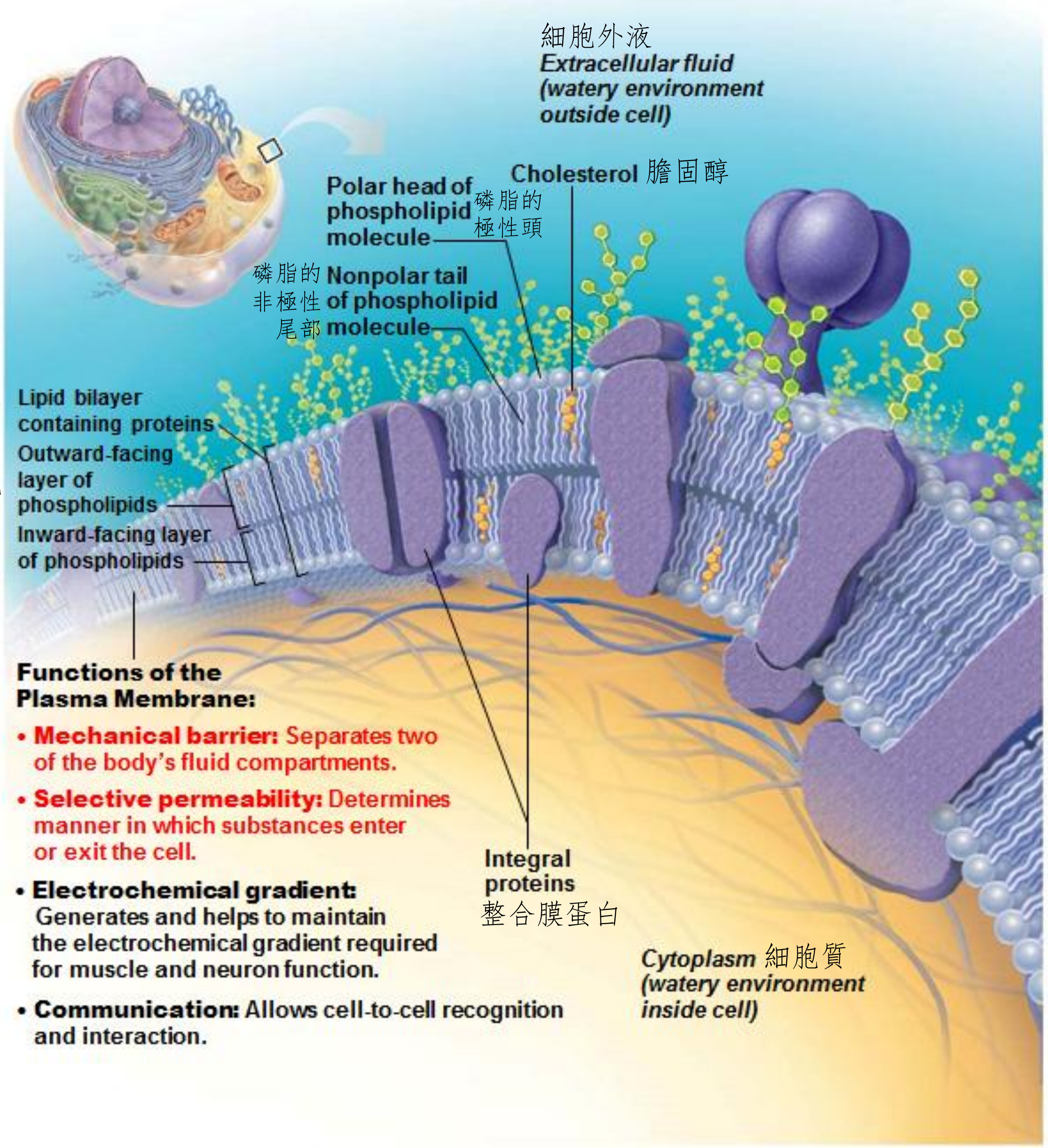
含蛋白質的雙層脂質

外層
磷脂

內層
磷脂

細胞膜的功能：

- 機械屏障：將身體的兩個液體隔室分開
- 選擇性滲透性：確定物質進入或離開細胞的方式
- 電化學梯度：產生並幫助維持肌肉和神經元功能所需的電化學梯度
- 通訊：允許細胞間識別和交互活動



Cell Organelles in a Typical Human Cell 典型人類細胞中的細胞器

- **Plasma membrane** (Cell membrane) **細胞膜**
- **Functions: 功能：**
  - **Separating cytoplasm** (intracellular materials) **and tissue fluid** (extracellular fluid) **分隔細胞質**(細胞內物質)**及組織液**(細胞外液)
  - **Controlling what enters and what leaves cell 控制進入和離開細胞的物質**
- Chemicals: 化學物質：
  - Consists of membrane lipids (**Phospholipids** and Cholesterols) that form a flexible **lipid bilayer** 由形成柔軟**雙層脂質結構**的膜脂質(**磷脂**和膽固醇)組成
  - Specialized **membrane proteins** float through this fluid membrane 特化的**膜蛋白**漂浮於流體膜之中
    - **Transporter proteins 轉運蛋白**
      - Channel proteins 通道蛋白, Carrier proteins 載體蛋白
    - **Receptor proteins 受體蛋白**

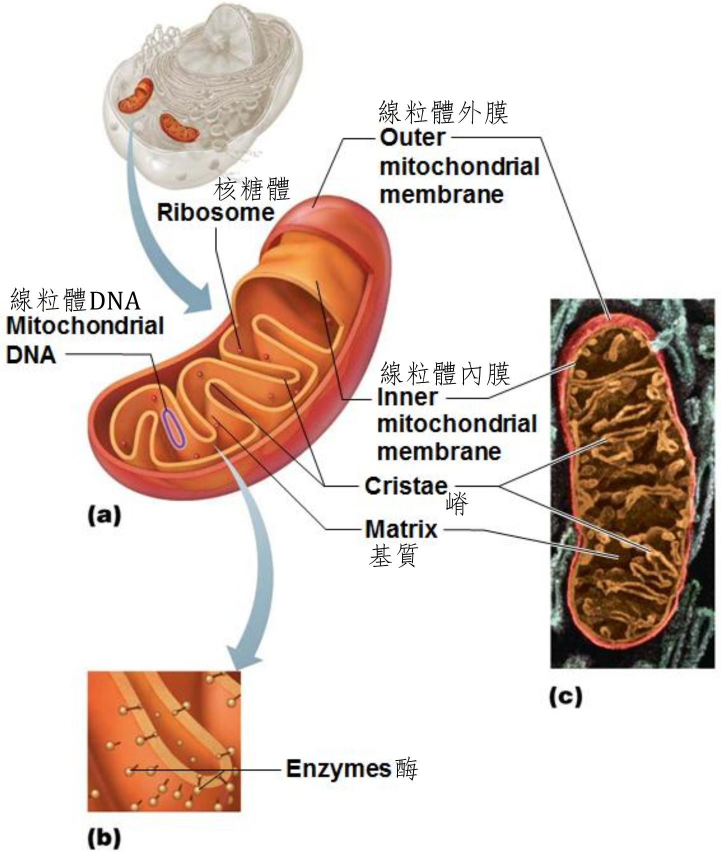
Cell Organelles in a Typical Human Cell
典型人類細胞中的細胞器

- **Cytoplasm 細胞質**
- All materials inside a cell 細胞內的所有物質
- **Cytosol** (Fluid) + **Cell organelles** (Cellular structures)
  胞漿 (液體) + **細胞器** (細胞結構)
- Important cell organelles: 重要的細胞器：
  - Nucleus (Nuclei), 細胞核
  - Mitochondrion (Mitochondria), 線粒體
  - Endoplasmic reticulum (Smooth / Rough), 內質網 (光滑/粗面)
  - Lysosome, 溶酶體
  - Golgi body 高爾基體

Cell Organelles in a Typical Human Cell
典型人類細胞中的細胞器

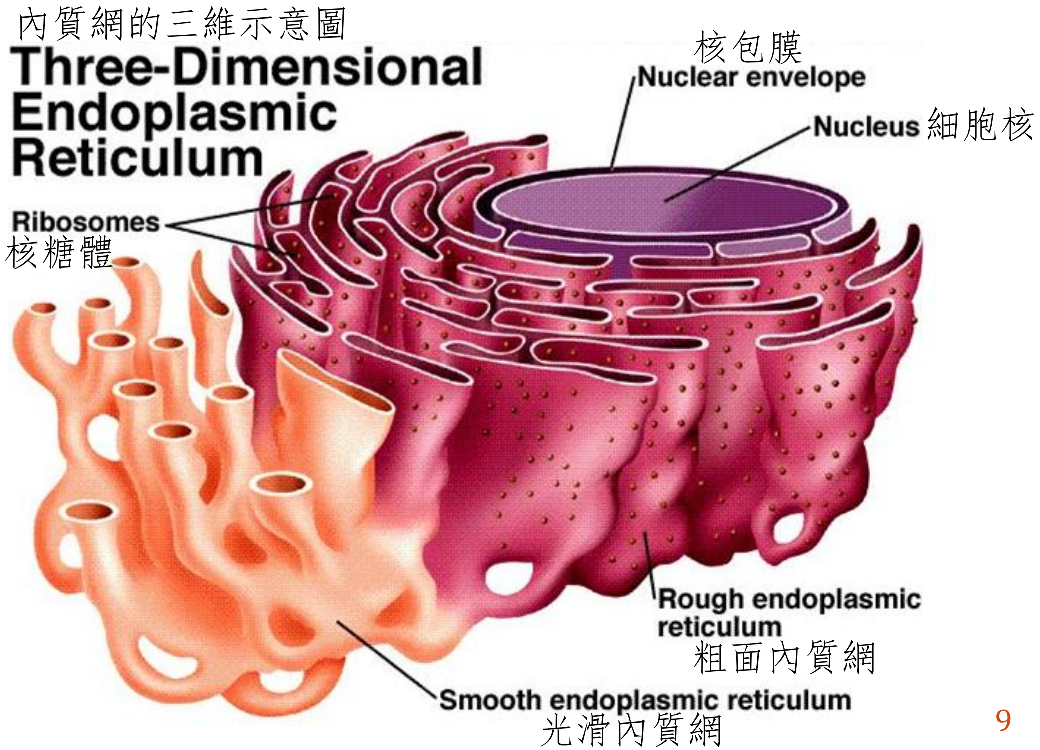
- **Mitochondrion** (Mitochondria) **線粒體**
- "**power plant**" of cell 細胞的「**發電廠**」
- Perform aerobic (oxygen-requiring) cellular respiration to **produce most of cell's energy**
  進行有氧(需要氧氣)細胞呼吸以**產生大部分細胞能量**
- Formed by 2 membranes 由2層膜形成

Which type of cell, fat cell or muscle cell, has more mitochondria in cytoplasm? Why?
哪種類型的細胞，脂肪細胞或肌肉細胞，細胞質中的線粒體更多？為什麼？



Cell Organelles in a Typical Human Cell
典型人類細胞中的細胞器

- **Endoplasmic Reticulum (ER)內質網**
- Membranous organelles **extending from the membrane of nucleus**
  從核包膜延伸的膜狀細胞器
- Locate near the nucleus
  位於細胞核附近
- 2 types:2 種類型：
  - **Smooth ER 光滑內質網**
    (without ribosome沒有核糖體)
  - **Rough ER 粗面內質網**
    (with ribosome有核糖體)



Cell Organelles in a Typical Human Cell
典型人類細胞中的細胞器

- **Rough ER 粗面內質網**
  - External surface appears rough because of the **attached ribosomes** 由於**附著的核糖體**，外表面顯得粗糙
  - **Site of synthesis of proteins and phospholipids 合成蛋白質和磷脂的位置**
  - Final protein is enclosed in vesicle and sent to Golgi body for further processing 合成的蛋白質被包裹在囊泡中並送至高爾基體進行加工

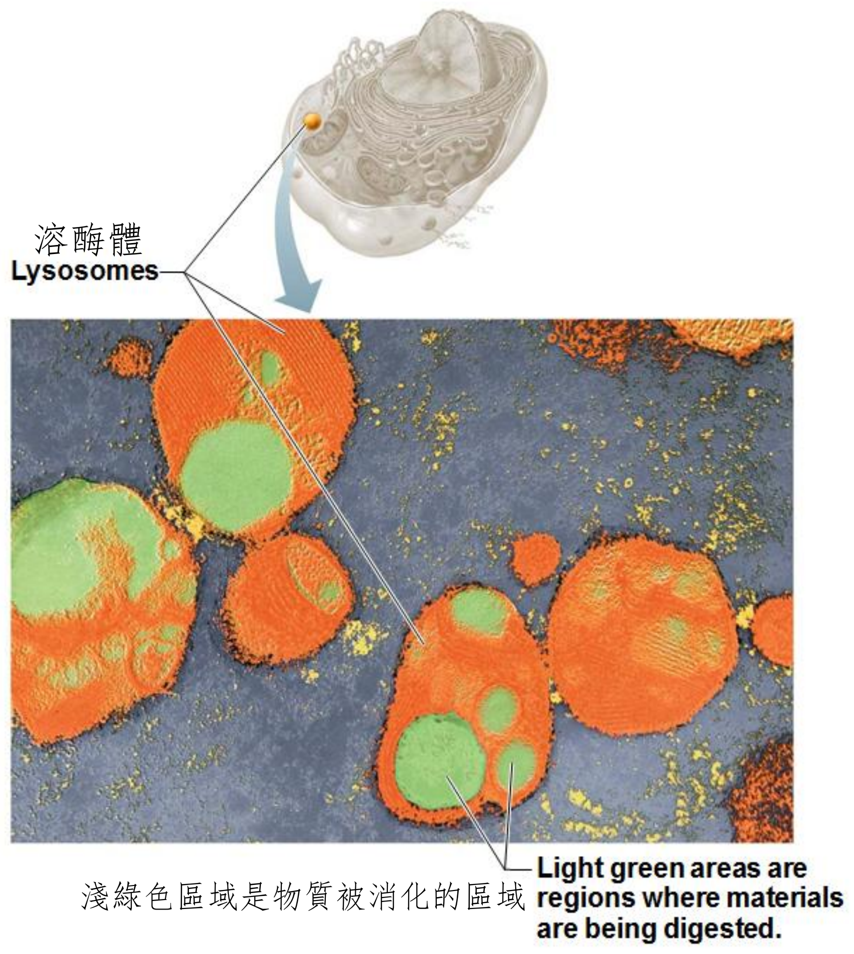
- **Smooth ER 光滑內質網**
  - Cholesterol and steroid-based hormone synthesis 合成膽固醇和類固醇激素
  - Converting of glycogen to free glucose 將糖原轉化為遊離葡萄糖
  - Storage and release of calcium ions 儲存和釋放鈣離子

Cell Organelles in a Typical Human Cell
典型人類細胞中的細胞器

- **Golgi body** (Golgi complex / Golgi apparatus)**高爾基體**
- Stacked and flattened membranous organelles 堆疊和扁平的膜狀細胞器
- **Modifies, sorts, and packages proteins and lipids received from rough ER 修改、分類和包裝從粗面內質網接收的蛋白質和脂質**
  - Pack into vesicles to release outside cell
    包裝到囊泡中以釋放到細胞外
  - Pack into vesicles to fuse with cell membrane
    包裝到囊泡中以與細胞膜融合
  - Pack into lysosomes
    包裝到溶酶體中

Cell Organelles in a Typical Human Cell
典型人類細胞中的細胞器

- **Lysosome 溶酶體**
  - Spherical membranous organelle **containing digestive enzymes 含有消化酶**的球形膜細胞器
  - Digest ingested bacteria, viruses, and toxins
    消化攝入的細菌、病毒和毒素
  - Degrade damaged organelles
    降解受損的細胞器



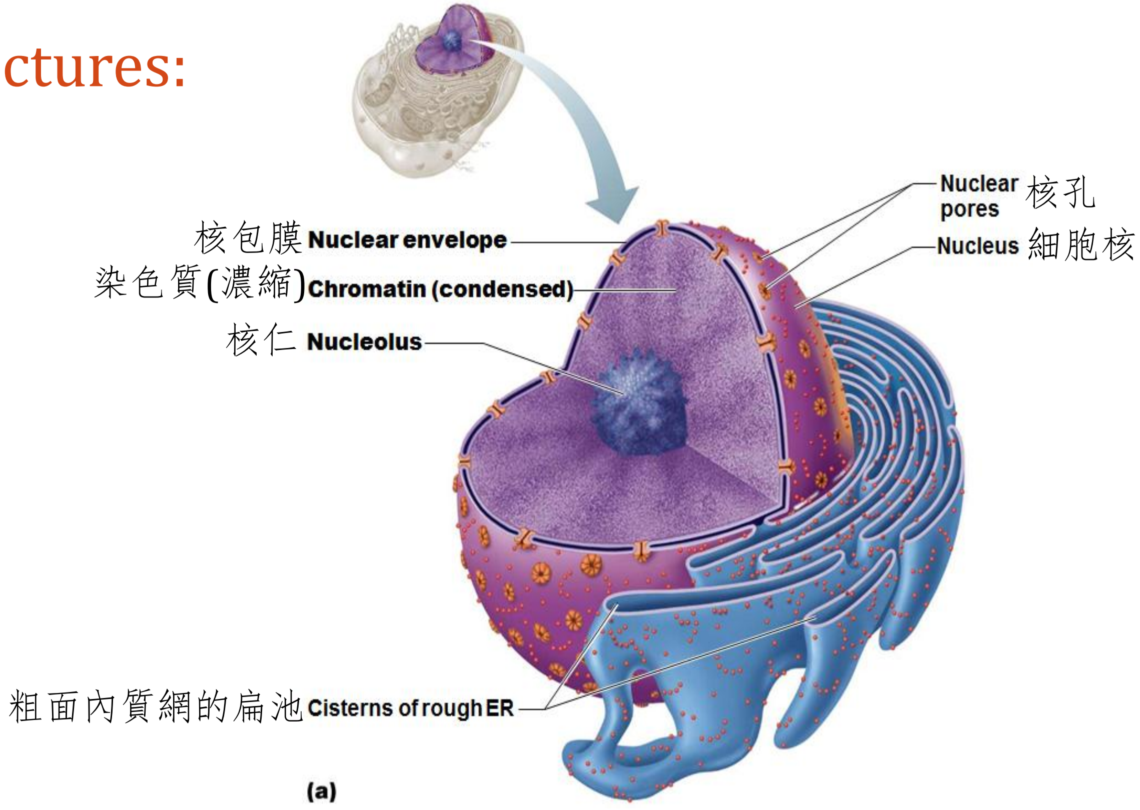
Cell Organelles in a Typical Human Cell
典型人類細胞中的細胞器

- **Nucleus 細胞核**
- Largest organelle最大的細胞器
- **Contains the genetic information** (Chromosomes) for synthesis of nearly all proteins 包含合成幾乎所有蛋白質的遺傳資訊 (染色體)
  - Responds to signals to control the kinds and amounts of proteins to be produced 回應各種信號以控制要產生的蛋白質的種類和數量

- Most cells are **uninucleate** (one nucleus) 大多數細胞是**單核**細胞(一個細胞核)
- Skeletal muscle cells, some bone cells, and some liver cells are **multinucleate** (many nuclei) 骨骼肌細胞、一些骨細胞和一些肝細胞是**多核**的(許多細胞核)
- Mature red blood cells are **anucleate** (no nucleus) 成熟紅血球細胞為**無核**的(沒有細胞核)

Cell Organelles in a Typical Human Cell 典型人類細胞中的細胞器

- The nucleus has three main structures:
  細胞核有三個主要結構:



- **Nuclear membrane核膜**
  (Nuclear envelope核包膜)
- **Nucleolus** (Nucleoli) **核仁**
- Chromatin染色質
  (Chromosome染色體)

Cell Organelles in a Typical Human Cell
典型人類細胞中的細胞器

- **Nuclear membrane 核膜**
  - Double-membrane barrier that encloses the substances inside nucleus
    將物質包裹在細胞核內的雙膜屏障
  - Outer layer is continuous with rough ER
    外層與粗面內質網連接
  - **Nuclear pores** allow substances to pass into and out of nucleus
    核孔允許物質進出細胞核

- **Nucleolus 核仁**
  - Dark-staining spherical body within nucleus 呈深色的細胞核內球體
  - Involved in **ribosome synthesis** 參與**生成核糖體**

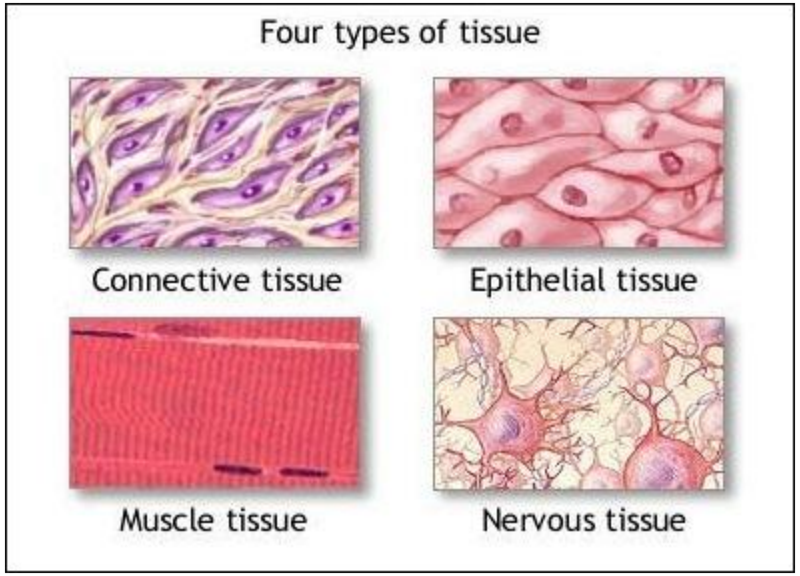
Summary: Major functions of different cell parts
摘要：不同細胞部位的主要功能

Cell Part 細胞部位	Major Function(s) 主要功能
Cell membrane細胞膜	• Separate cytoplasm from extracellular fluid 將細胞質與細胞外液分隔 • Control substances entering or leaving the cell 控制物質進出細胞
Smooth ER 光滑ER	• Metabolism of lipid and carbohydrate脂質和碳水化合物的代謝
Rough ER 粗面ER	• Site of protein synthesis (ribosome) 合成蛋白質的位置（核糖體） • Production of phospholipids 生產磷脂
Golgi body 高爾基體	• Modify, sort, and package substances received from rough ER 修改、分類和包裝從粗面ER接收的物質
Mitochondrion 線粒體	• Produce cellular energy 產生細胞能量
Lysosome 溶酶體	• Digest foreign substances and damaged organelles 消化外來異物和受損的細胞器
Nucleus 細胞核	• Carry genetic information to control all cell activities 攜帶遺傳資訊以控制所有細胞活動
Nuclear membrane 核膜	• Control substances entering or leaving nucleus 控制物質進入或離開細胞核
Nucleolus 核仁	• Site of ribosome synthesis 合成核糖體的位置

Primary Tissues in Human Body
人體的基本組織

- **Tissue 組織**
- Group of cells that are similar in structure and perform common or related function 結構相似並執行共同或相關功能的細胞群

- Four basic tissue types: 四種基本組織類型:
- **Epithelial tissue 上皮組織**
- **Connective tissue 結締組織**
- **Muscle tissue 肌肉組織**
- **Nervous tissue 神經組織**



Primary Tissues in Human Body 人體的基本組織

- **Epithelial tissue 上皮組織** (epithelium上皮)
- a sheet of cells that covers body surfaces or cavities 覆蓋體表面或空腔的一層細胞

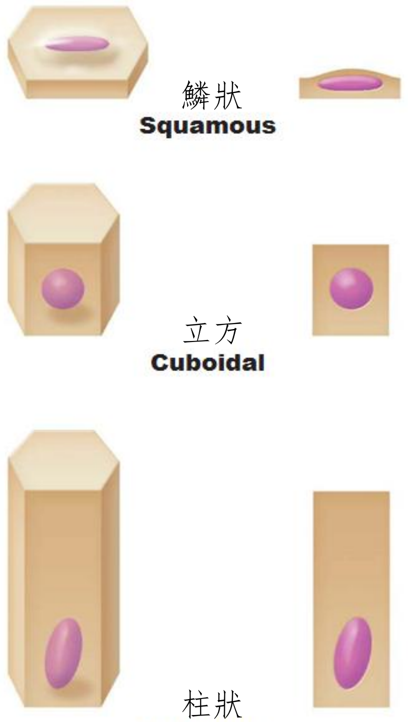
- Two main forms:兩種主要形式：
- **Covering and lining epithelium**: covering external and internal surfaces
  覆蓋上皮：覆蓋人體外表面和內表面
  - example: skin, lining of oral cavity 例子：皮膚、口腔黏膜
- **Glandular epithelium**: Secretory tissue in glands
  腺上皮：腺體中的分泌組織
  - example: salivary glands 例子：唾液腺

- Main functions:主要功能：
- **Protection of underlying parts 保護底下之部份**
- **Absorption 吸收**
- **Secretion 分泌**
- **Sensory reception 感官感受**

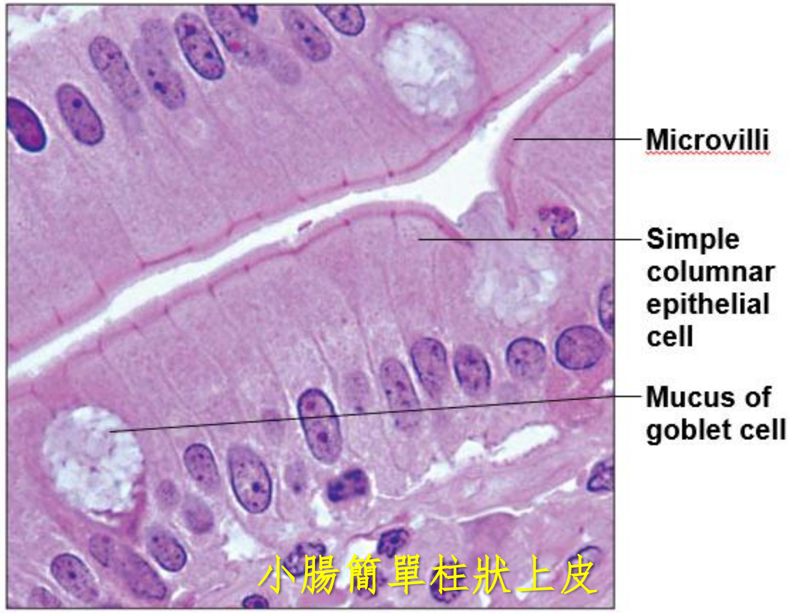
Primary Tissues in Human Body
人體的基本組織

- **Lining epithelium** can be divided into 3 types based on their cell shape:
  覆蓋上皮根據其細胞形狀可分為 3 種類型：
- Squamous epithelium 鱗狀上皮
  - E.g. epidermis of skin
    例如皮膚表皮
- Columnar epithelium 柱狀上皮
  - E.g. lining of digestive tract
    例如消化道內壁
- Cuboidal epithelium 立方上皮
  - E.g. ducts of sweat glands
    例如汗腺導管

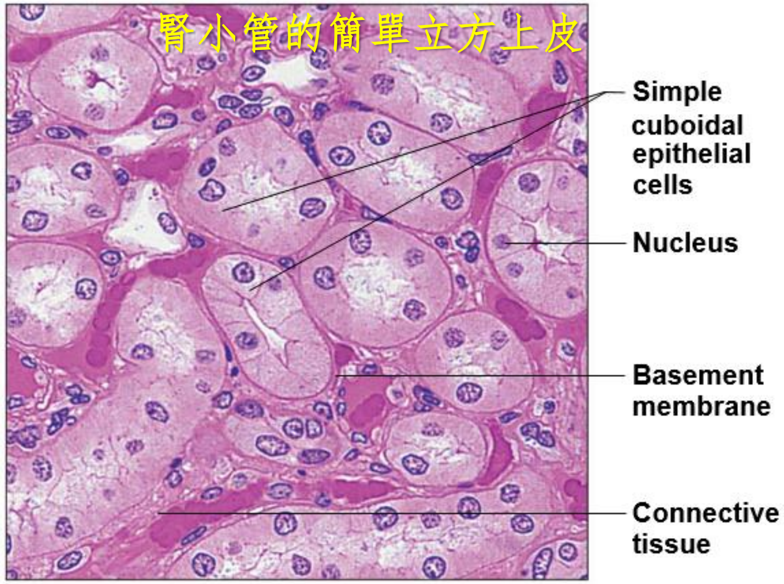
- All types allow **constant loss and renewal of cells**
  所有類型都允許**細胞的恆常流失和更新**



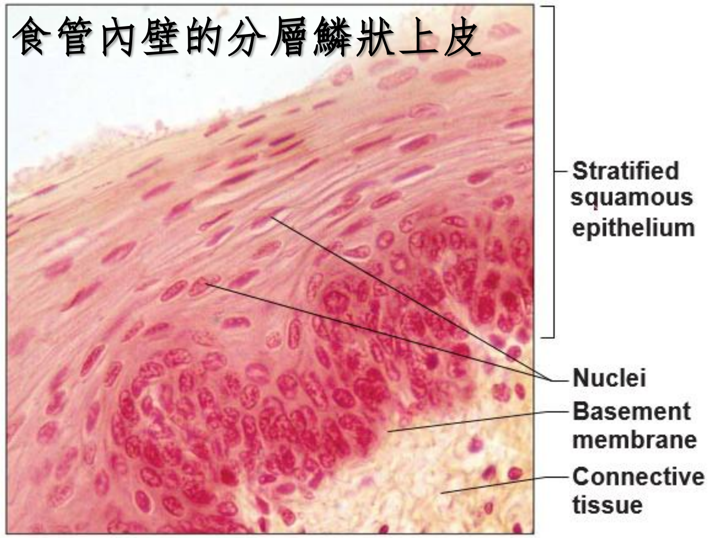
(b) Classification based on cell shape.
根據細胞形狀分類



Photomicrograph: Simple columnar epithelium of the small intestine



Photomicrograph: Simple cuboidal epithelium in kidney tubules



Photomicrograph: Stratified squamous epithelium lining the esophagus

Simple epithelium 簡單上皮:
single layer thick 單層細胞厚度

Stratified epithelium 分層上皮:
two or more layers thick 兩層或多層細胞厚度

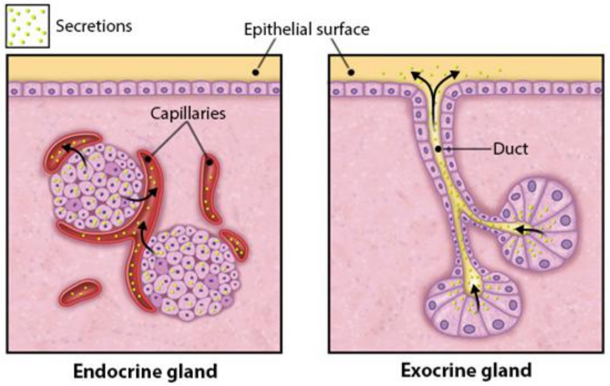
Primary Tissues in Human Body
人體的基本組織

- **Glandular epithelium腺上皮** (Gland腺體)
  - One or more cells that produces and secretes chemicals out of cell
    一個或多個細胞產生化學物質並分泌到細胞外

- Classification:分類：
  - **Endocrine gland 內分泌腺**:
    - secreting hormones 分泌激素
  - **Exocrine gland 外分泌腺**:
    - secreting non-hormone chemicals 分泌非激素化學物質

Primary Tissues in Human Body
人體的基本組織

	Endocrine gland內分泌腺	***Exocrine gland外分泌腺***
Structure 結構	**Ductless 無導管**	With duct 有導管
Secretion 分泌	Hormones 激素	Non-hormone chemicals 非激素化學物質
Destination of secretion 分泌物的目的地	Blood 血	Body surface or body cavity 體表 或 體腔
Example 例子	Adrenal gland is an endocrine gland that secretes adrenaline into blood. 腎上腺是將腎上腺素分泌到血液中的內分泌腺。	Salivary gland is an exocrine gland that secretes saliva into oral cavity. 唾液腺是將唾液分泌到口腔的外分泌腺。



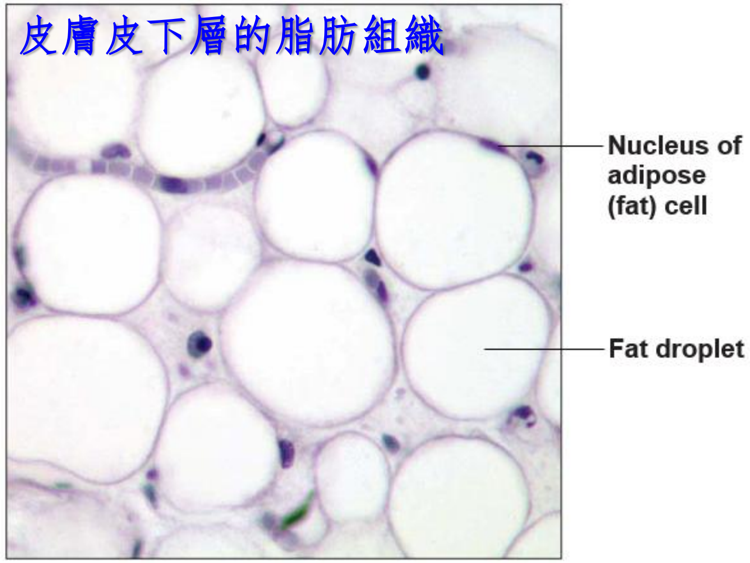
內分泌腺　外分泌腺

Primary Tissues in Human Body
人體的基本組織

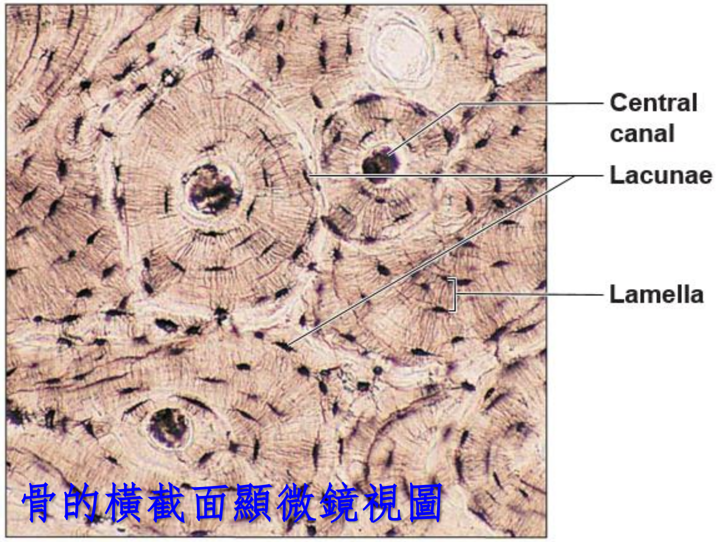
- **Connective tissue 結締組織**
  - Most abundant and widely distributed primary tissues
    最豐富和分佈最廣泛的基本組織

- Main classes 主要類別
  - **Connective tissue proper 固有結締組織**
  - **Cartilage 軟骨**
  - **Bone 骨**
  - **Blood 血**
  - **Adipose tissue (Fat tissue) 脂肪組織**

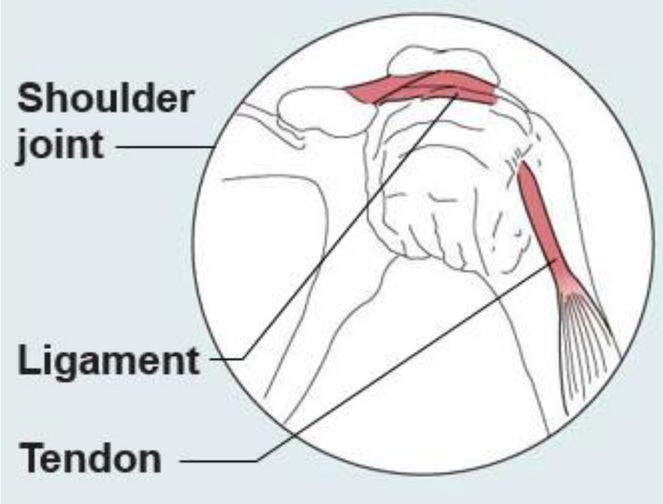
Type of Connective Tissue ***結締組織類型***	***Examples 例子***	***Main Function(s) 主要功能***
Connective tissue proper 固有結締組織	Dermis of skin 皮膚真皮 Tendons 肌腱 Ligaments 韌帶	➢ Binding different body parts 連接不同的身體部位 ➢ Protection 保護
Cartilage 軟骨	Articular cartilages 關節軟骨	➢ Cushion and protect body structures 緩衝和保護身體結構
Bone 骨	Skull 顱骨 Vertebral column 脊柱 Long bone 長骨 Flat bone 扁平骨	➢ Support body framework 支撐人體框架 ➢ Produce blood cells 產生血細胞 ➢ Storage of mineral salts 儲存礦物鹽
Blood 血	---	➢ Carry substances to different body parts 將物質運輸到身體的不同部位 ➢ Defense against disease-causing agents 防禦致病因子
Adipose tissue 脂肪組織	Subcutaneous fat 皮下脂肪 Ectopic fat 異位脂肪	➢ Insulation 隔熱 ➢ Storage of nutrients 儲存營養物質



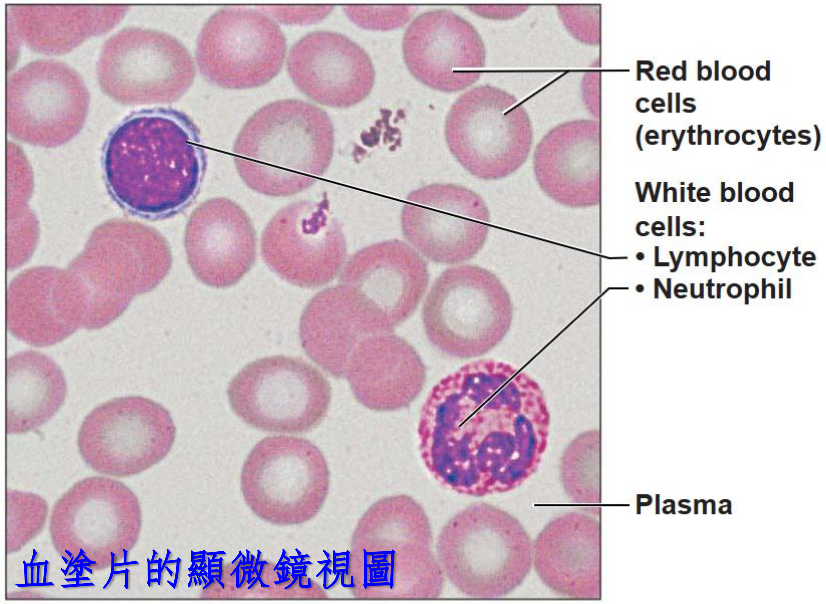
Photomicrograph: Adipose tissue from the subcutaneous layer under the skin



Photomicrograph: Cross-sectional view of bone



肩關節韌帶和肌腱



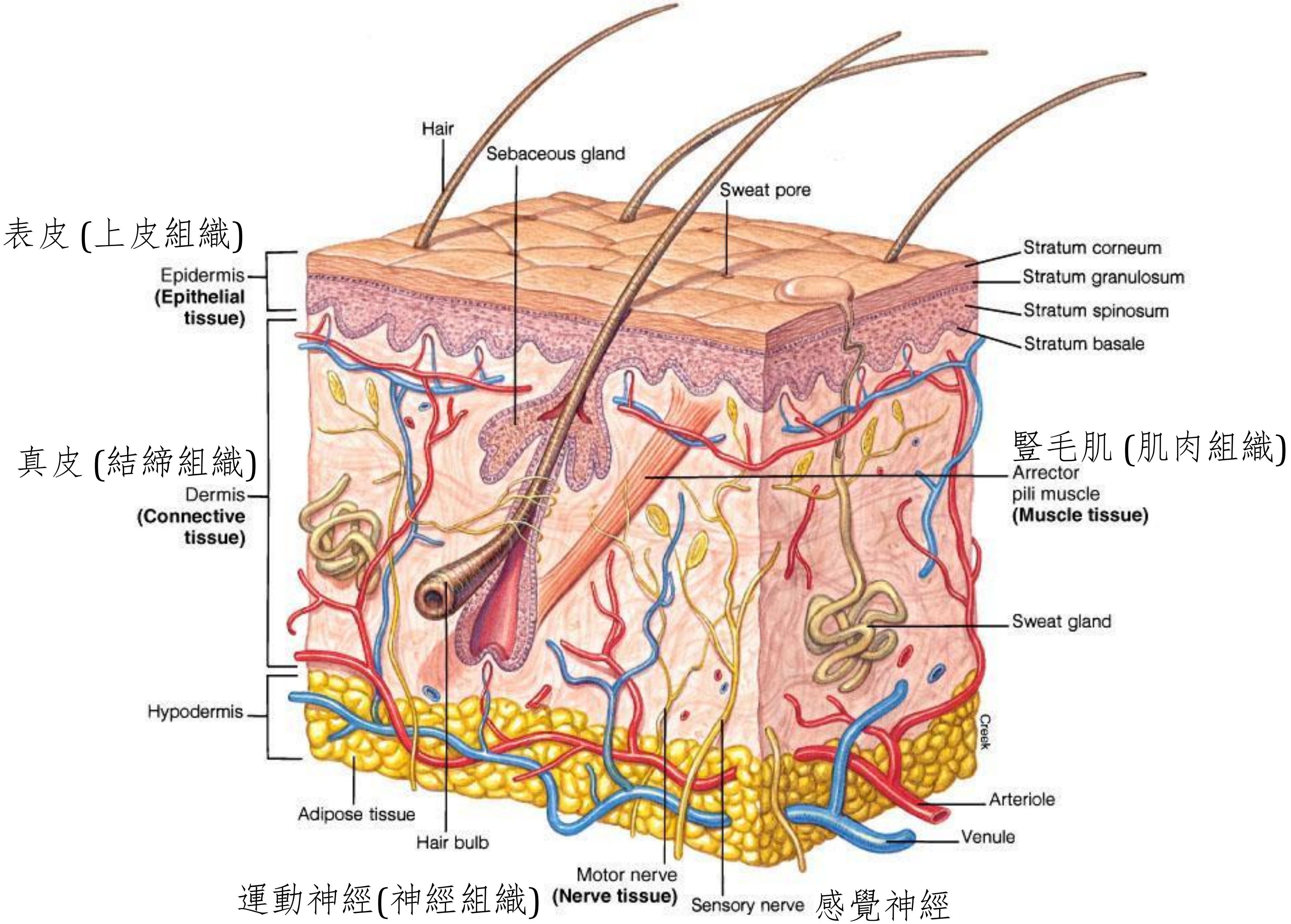
Photomicrograph: Smear of human blood

Primary Tissues in Human Body
人體的基本組織

- **Muscle tissue 肌肉組織**
  - **Responsible for most types of movement 負責人體大多數類型的活動**
  - Muscle cells have proteins that bring about **contraction**
    肌肉細胞具有導致**收縮**的蛋白質
  - Three types of muscle tissues:
    三種類型的肌肉組織：
    - **Skeletal muscle tissue 骨骼肌組織**
    - **Cardiac muscle tissue 心肌組織**
    - **Smooth muscle tissue 平滑肌組織**

Primary Tissues in Human Body
人體的基本組織

- **Nervous tissue 神經組織**
  - Main component of organs in nervous system (brain, spinal cord, nerves)
    神經系統器官 (腦、脊髓、神經)的主要成分
  - **Generate and transmit electrical signals to control body functions**
    生成和傳輸電信號以控制身體功能
  - Made up of two specialized cell types: 由兩種特殊的細胞類型組成：
    - **Neurons**: specialized nerve cells that generate and conduct nerve impulses
      神經元： 產生和傳導神經衝動的特殊神經細胞
    - **Supporting cells** that support, insulate, and protect neurons
      支援細胞支援、隔離和保護神經元


Hair
Sebaceous gland
Sweat pore
表皮 (上皮組織)
Epidermis
(Epithelial tissue)
Stratum corneum
Stratum granulosum
Stratum spinosum
Stratum basale
真皮 (結締組織)
Dermis
(Connective tissue)
豎毛肌 (肌肉組織)
Arrector pili muscle
(Muscle tissue)
Sweat gland
Hypodermis
Creek
Arteriole
Venule
Adipose tissue
Hair bulb
Motor nerve
(Nerve tissue)
運動神經(神經組織)
Sensory nerve
感覺神經

L2 Revision Exercise L2 複習練習

Screen the QR code below to complete all questions in this revision exercise.
掃瞄下面的二維碼以完成此複習練習中的所有問題。

References

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